BIPHMUN 2026

Background Guide

Committee: United Nations Environment Programme

Topic: Building Climate-Resilient Digital Infrastructure in Developing Economies

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Letter from the Chair

Dear Delegates,

Welcome to BIPHMUN 2026. My name is Noah Chen, and I am a sophomore studying at BIPH. I will serve as your Co-Chair along with Chair Mia Chen.

The United Nations Environment Programme (UNEP) is the leading global authority on the environment, working to address the triple planetary crisis of climate change, nature loss, and pollution. It serves as a central forum for setting globally influential policies and guidelines.

This committee's topic, Building Climate-Resilient Digital Infrastructure in Developing Economies, focuses not only on solving technical challenges but also on governance gaps and general environmental planning issues. Delegates are tasked with maneuvering within the feedback loops between Digital Infrastructure, Climate Change, and better Climate resilience, all the while keeping in mind the growing digital divide.

Delegates should come prepared and be ready to engage in meaningful, foresighted and professional discussions. We hope to have a wonderful time and hope to see you soon!

Sincerely,
Noah Chen

Introduction

The United Nations Environment Programme (UNEP) is a leading global organization for environment protections, with three main goals: climate change, the loss of nature

and pollution. Technology stands as an important factor with the UNEP's plan. UNEP is dedicated to protect the earth by implying technology tools, while ensuring the technology industry itself don't causing so much pollution.

When we discuss the topic "Building Climate-Resilient Digital Infrastructure in Developing Economies", we look at how climate change, economic, and technologies connect to each other. Every time we talk about the internet or "cloud service", we are actually talking about entities in the physical world: the global network is composed of massive data centers, signal towers, and thousands of miles of cables.

This reliance on physical connectivity created a geographic bias that is now coming back to us as a threat. During the time of the telecom boom back in the 1990s, the whole world was competing with the physical foundational infrastructure. Developers place rapid expansion and economic efficiency on top of long-term resilience to the

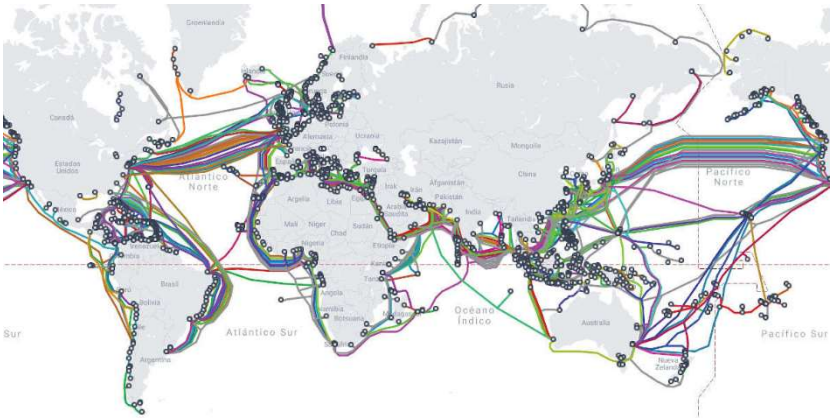

Fig 1: Global Map of Submarine Cable Networks
(Undersea Cables' Vulnerability: A Hidden Network of Vital Connectivity, n.d.)

environment. Thus, nearly ten thousand kilometers of fiber cables and landing stations were installed along the coastlines and highways for chasing convenience. This mass scale of global deployment was built on top of a dangerous assumption: the climate is always stable.

History of the Issue

The development of global digital infrastructure is closely tied to the expansion of the modern internet beginning in the late twentieth century. During the 1980s and 1990s, rapid technological advancement and the liberalization of telecommunications markets triggered

a global surge in investment in fiber-optic cables, satellite networks, and telecommunications towers. Governments and private companies focused primarily on expanding connectivity as quickly and efficiently as possible, often prioritizing economic feasibility and geographic convenience over long-term environmental considerations. As a result, a large portion of global digital infrastructure—including submarine cable landing stations, data centers, and telecommunications hubs—was built in coastal areas, along transportation corridors, or near major population centers.

At the time, climate change was not widely incorporated into infrastructure planning. The placement of digital systems assumed relatively stable environmental conditions, meaning that issues such as sea-level rise, extreme heat, and intensified storms were rarely considered during construction. This assumption has become increasingly problematic as climate change accelerates. Many key components of the global internet are now located in areas exposed to rising sea levels, coastal flooding, and extreme weather events.

In developing economies, the risks are particularly significant. These countries are simultaneously experiencing rapid growth in digital connectivity while facing heightened vulnerability to climate hazards. As governments expand broadband access, mobile networks, and data infrastructure to support economic development, the challenge becomes ensuring that these systems can operate reliably during climate-related disasters. Extreme heat can disrupt data centers and telecommunications equipment, flooding can damage cable landing stations and network facilities, and storms can destroy signal towers that provide connectivity to rural populations.

At the same time, the digital sector itself contributes to environmental pressures. Data centers, network infrastructure, and device manufacturing require large amounts of energy and raw materials. As global demand for cloud services, artificial intelligence, and digital platforms grows, the environmental footprint of the information and communication technology (ICT) sector continues to expand. Estimates suggest that ICT systems account for a growing share of global greenhouse gas emissions and energy consumption.

These interconnected challenges have gradually drawn the attention of policymakers,

researchers, and international organizations. Over the past decade, the conversation has shifted from simply expanding digital connectivity toward ensuring that digital infrastructure is resilient, sustainable, and inclusive. Increasingly, governments and international institutions recognize that digital infrastructure must be designed to withstand climate risks while also supporting climate adaptation, disaster preparedness, and environmental monitoring. This evolving perspective has shaped many of the international initiatives outlined in the following section.

Today, this physical technology in developing countries is in danger. With the rise of sea level and severe storms can flood the buildings where ocean cables connect to the land. Moreover, extreme heatwaves can cause data centers to overheat and shut down. As a reference, during the terrible 2022 floods in Pakistan, internet cables were destroyed, leaving millions of people without access to emergency warnings or digital cash aid.

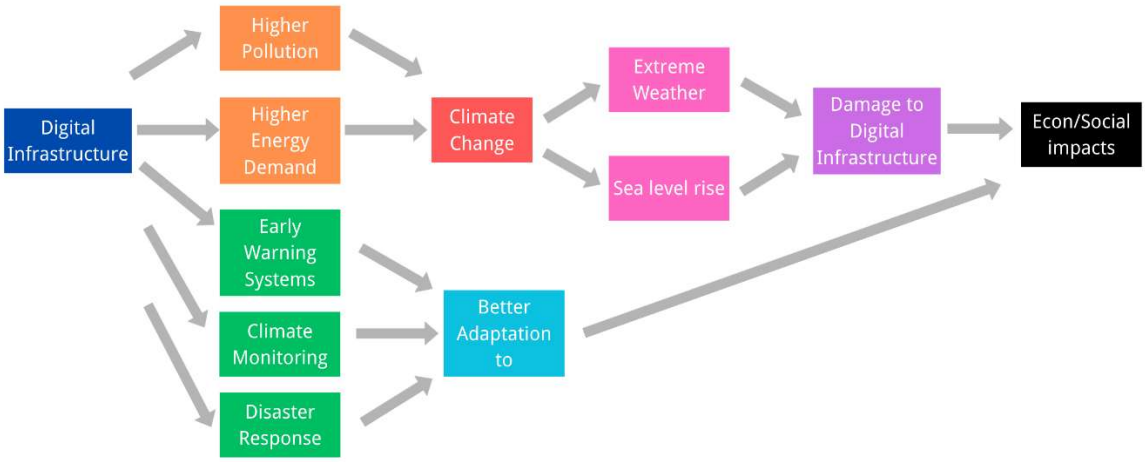

Fig 2. Diagram of Digital Infrastructure and Climate feedback loop. Created using canva.com

Thus, it is very important to recognize this issue not as mere technological challenges, but also as a **challenge in governance, environmental policy planning, and digital development.** Although it is not the biggest climate emitter, it is fast growing. Studies show the ICT sector is emitting around 1.5% to 4% of global GHG emissions. (Rajani, 2024). Digital infrastructure itself creates more threats to the environment and the climate, which in turn leads to more threatening environments (e.g. climate hazards and sea level rise) for digital infrastructure to thrive. This leads to disastrous outcomes in economic and social sectors. But on the positive side, more developed digital infrastructure can be used in early warning systems, climate monitoring, and better disaster handling, helping societies adapt to climate change.

Thus, debates should revolve around three competing goals: digital development, climate resilience, and environmental sustainability. Delegates must consider how developing economies can expand digital connectivity—through broadband networks, data centers, and telecommunications systems—while ensuring that these infrastructures are capable of withstanding intensifying climate hazards such as floods, storms, and heatwaves. At the same time, the environmental footprint of the rapidly growing digital sector must be addressed through sustainable energy use, efficient design, and responsible lifecycle management of technological equipment. Within this context, the international community faces a complex policy challenge: designing governance frameworks, financing mechanisms, and cooperative initiatives that allow developing economies to build resilient and sustainable digital systems without slowing economic progress or deepening the global digital divide. The task of this committee, therefore, is to explore solutions that balance these three objectives and support a stable, inclusive, and environmentally responsible digital future.

Definition of Key Terms

Digital Public Infrastructure

Digital systems and platforms that enable the delivery of services, facilitate data exchange, and support digital governance across various sectors.

Climate Resilience

Capacity of people, livelihoods, businesses, economies, and environmental systems to anticipate, manage, and thrive amid climate risks, achieved through protective actions like early warnings, green jobs, climate-informed investment, and the safeguarding of natural ecosystems as a primary defense. (UNFCCC, n.d.)

Nature based Solutions

Actions to protect, conserve, restore, sustainably use, and manage natural or modified terrestrial, freshwater, coastal, and marine ecosystems, which address social, economic, and environmental challenges effectively and adaptively, while simultaneously providing human well-being, ecosystem services, resilience, and biodiversity benefits. (UNEA, 2022)

Digital Divide

The unequal distribution of access to Information and Communication Technologies (ICT), the skills to use them, and the ability to derive income or benefits from them across different countries, social groups, and hierarchies. This divide is often analyzed in three levels: (1) inequality of ownership and access to ICTs; (2) inequality in the skills to use ICTs; and (3) inequality in the ability to create income or benefits by using ICTs. (Olgun, 2024)

Early Warning Systems

An integrated system of hazard monitoring, forecasting and prediction, disaster risk assessment, and communication and preparedness activities that enable individuals, communities, governments, businesses, and others to take timely action to reduce disaster risks in advance of hazardous events . These are extensive networks that use expert and disaster risk knowledge to provide timely warnings and minimize the loss of life and livelihoods. (United Nations Office for Disaster Risk Reduction – Regional Office for the Americas and the Caribbean, 2023)

Green ICT

The practice and study of designing, manufacturing, using, and disposing of computers, servers, and associated subsystems—such as monitors, printers, storage devices, and networking and communications systems—efficiently and effectively with minimal to no environmental impact . The notion has been envisioned to reduce, and possibly eliminate, the environmental issues caused by ICT through sustainable designs and approaches. (Butt et al., 2020)

Debt for Climate Swaps

Partial debt relief operations, conditional on a debtor country's commitment to undertake climate-related investments. These are financial mechanisms where a portion of a country's debt is reduced or restructured in exchange for the government using the resulting savings to fund domestic climate adaptation, mitigation, or other environmental projects. (Chamon et al., 2022)

Past International Actions

Year	Milestone	Core Objective / Outcome
2015	Adoption of the 2030 Agenda (SDGs) & Sendai Framework	Established resilient infrastructure (SDG 9) as essential for sustainable development and formalized proactive disaster risk reduction.
2021	Creation of Coalition for Digital Environmental Sustainability (CODES)	Multi-stakeholder alliance formed by UNEP, UNDP, and ITU to align digital transformation with environmental sustainability.
2022	Launch of CODES Action Plan for a Sustainable Planet	Proposed 3 systemic shifts and 9 Impact Initiatives to mitigate the ICT sector's climate footprint and boost digital adaptation.
2023	Launch of the Green Digital Action (GDA) GHG pillar at COP28	ITU and partners launched the GHG Track under Green Digital Action to reduce ICT Sector GHG emissions through collaboration, robust accountability and transparent reporting (Rajani, 2024)
2024	Adoption of the Global Digital Compact (GDC)	UN framework committing to close the digital divide, promote safe DPI, and ensure lifecycle sustainability of tech.
2024	COP29 Declaration on Green Digital Action	First formal COP declaration prioritizing the climate-proofing of digital infrastructure and reducing tech emissions.
2025	Sevilla Commitment (FFD4)	Global agreement to scale up development financing specifically targeted at resilient digital infrastructure and domestic capacity.

Table 1. Timeline

International attention to the intersection of digital infrastructure, climate resilience, and sustainable development has gradually expanded over the past decade. Early global frameworks such as the 2030 Agenda for Sustainable Development and the Sendai Framework for Disaster Risk Reduction, both adopted in 2015, emphasized the importance of resilient infrastructure and proactive disaster risk management. While these agreements were not initially focused on digital systems specifically, they established the broader

policy foundation for integrating climate resilience into infrastructure planning.

As digital technologies became increasingly central to economic and social systems, international organizations began to address the environmental impacts of the ICT sector more directly. In 2021, the Coalition for Digital Environmental Sustainability (CODES) was established to align digital transformation with environmental goals. Its 2022 Action Plan further proposed systemic changes to reduce the environmental footprint of digital technologies while leveraging digital tools for climate adaptation and monitoring.

More recent initiatives have expanded global coordination in this area. Programs such as Green Digital Action promote collaboration between governments, industry, and international institutions to reduce ICT-related emissions and improve transparency in sustainability reporting. Meanwhile, frameworks such as the Global Digital Compact emphasize closing the digital divide and ensuring responsible digital development, particularly in developing economies.

Together, these initiatives reflect a growing international consensus that digital infrastructure must be developed in ways that are both environmentally sustainable and resilient to climate risks.

Bloc Stances

Global South and Developing Countries

Representing the Global South, this bloc demands equitable technology transfer and unconditional climate financing from developed nations to build resilient digital infrastructure. They fiercely advocate for digital sovereignty, localized data centers, and open-source systems to avoid Western tech monopolies. Furthermore, they strongly oppose unilateral trade restrictions disguised as environmental policies that could hinder their economic growth and digital transformation.

Small Island Developing Countries (SIDCs)

For these vulnerable nations, climate resilient digital infrastructure ensures existential survival. Highly dependent on fragile subsea cables, they fiercely demand redundant connectivity and satellite backups. They urge the international community to adopt vulnerability indexes for concessionary financing. Furthermore, they heavily promote innovative fiscal tools like debt relief swaps to fund vital network upgrades without worsening their crippling sovereign debt burdens.

European Union (EU)

The European Union champions a strictly regulated approach to green and digital transitions, leveraging its massive market power to establish global standards. Through its transparent infrastructure investment strategies, the bloc prioritizes human rights, strict data privacy, and environmental sustainability. Strategically, this normative framework offers an alternative development model while actively maneuvering to reduce critical supply chain dependencies on geopolitical rivals.

African Union (AU)

Guided by unified regional strategies, this bloc views resilient digital connectivity as the primary catalyst to leapfrog traditional industrialization. They prioritize expanding rural broadband, deploying green energy grids, and fostering regional data sharing. They insist international partnerships must move beyond historical extraction models, demanding investments that respect state fragility while actively funding youth skills and vibrant local digital technology ecosystems.

United States and Western Allies

Driven by commercial interests and urgent national security imperatives, this allied bloc views telecommunications networks and data centers as critical geopolitical assets. They mobilize massive private capital to fund secure global connectivity while aggressively blocking vendors deemed high risk. Ultimately, their stance promotes cloud resilience while cementing the continued market dominance of major Western hyperscale technology corporations around the world.

People's Republic of China

Promoting its massive Digital Silk Road, China positions itself as a resourceful and supportive partner for some developing nations. By offering highly affordable network infrastructure and smart city technologies, they seek to effectively counterbalance Western hegemony. Furthermore, they strictly emphasize state sovereignty over internet governance while increasingly integrating vital green energy solutions into their rapidly expanding portfolio of overseas digital infrastructure investment projects.

Potential Solutions

Potential solutions can be categorized into infrastructure planning, regulation and governance, and financial and risk-sharing mechanisms.

I. Infrastructure Planning and Physical Resilience

This is the direct confrontation to climate change's effects on digital infrastructure. These are policies that change how and where digital infrastructure is built to reduce climate vulnerability. Examples could include Climate Risk Zoning for Digital Infrastructure, including restriction on construction of data centers and cable landing solutions in high-risk coastal or flood zones. As we can take inspiration from the Codon to Amino Acid mapping, redundancy can help avoid single points of failure, which could be applied to multiple submarine cable connections or hybrid systems (cable + satellites). Additionally, data centers could be restricted to climate-stable regions and connecting neighboring countries through cross-border fiber networks. However, high construction and operational costs could be a burden for countries, and the restrictions in construction may slow digital expansion in poorer regions. Cross-border infrastructure, on the other hand, may raise data sovereignty concerns.

II. Financial and Risk-sharing Mechanisms

In the case of a catastrophe actually happening, we could take inspiration from joint-stock companies that we have learnt in history, to address the financing gap for resilient digital infrastructure. This could include international insurance systems to cover damage to critical digital infrastructure during climate disasters. As mentioned previously, debt-for-climate swaps (i.e. debt given to help with investments in resilient

digital systems) could be useful. However, wealthier countries may resist financing such mechanisms. The over reliance on compensation may create a moral hazard (a situation where one party takes excessive risk because they do not have to bear the costs of the risk), as occurs in almost all insurance contracts, which may either make insurance contracts hard to arrange without supervision (which creates its own issues), or pose concerns for the *lack* of supervision given an insurance contract, as with the case of the over-expansion of Mortgage-Backed Securities in 2008.

III. Environmental Regulation of the ICT Sector and Governance improvements

Policies aiming at reducing the environmental footprint of digital infrastructure itself can take the form of regulations. There could be mandates or incentives for renewable energy usage and energy-efficient cooling systems. In the electronic components' lifecycle, we can set requirements in recycling to reduce E-waste. Open architecture systems, such as Open RAN, can allow for more efficient and transparent telecom technologies. The telecom industry could be regulated. For example, Climate-linked Spectrum licensing includes telecom operators to meet resilience standards (e.g. storm, backup) to receive or renew spectrum licenses, and they can be embedded in national telecommunications regulations. However, strict regulations and high compliance costs may discourage private investment and lay the burden on government-funded efforts. Further, renewable energy infrastructure may not exist in many developing countries.

Questions to Answer

- Should climate-resilient digital infrastructure prioritize redundancy and resilience even if it slows digital expansion in developing economies?
- Who should finance climate-proof digital infrastructure — developing states, private tech companies, or developed nations responsible for historic emissions?
- How can developing countries maintain digital sovereignty while relying on foreign technology companies for resilient infrastructure?

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